

A sealed autonomous prospective benchmark for PubMed/MeSH link-emergence ranking

Tang Vu

Affiliation: **TO BE SUPPLIED BEFORE SUBMISSION**

ORCID: **TO BE SUPPLIED BEFORE SUBMISSION**

Correspondence: **TO BE SUPPLIED BEFORE SUBMISSION**

Protocol version 1; sealed August 2026

Abstract

Literature-based discovery methods are often evaluated retrospectively on a small set of known examples, creating opportunities for case selection, parameter adaptation, and hindsight leakage. This protocol defines a no-human prospective benchmark for one narrower task: ranking pairs of Medical Subject Headings (MeSH) that will later acquire direct co-indexing links in complete annual PubMed baselines. The 2026 T0 source, exhaustive score-free candidate universe, metric contract, and predictions were hash-sealed before any T1 outcome source existed. T0 contains 39,994,988 distinct PubMed identifiers and 31,110 MeSH descriptors. Deterministic eligibility rules produced 7,310,895 exact-zero candidate pairs. A fixed-point Adamic–Adar score and five baselines were then computed for every candidate and placed in immutable total orders. An automated watcher accepts only sequential official 2027–2029 PubMed and MeSH release identities. Registered outcome, integrity, power, performance, and structural-diagnostic gates resolve mechanically to pass, fail, or abstain after the three-release window. No prospective outcome currently exists: the benchmark is `not_ready`, the formula is unvalidated, and validated knowledge-gap pairs number zero. Even a future pass would validate only PubMed/MeSH link-emergence ranking, not knowledge-gap detection, scientific novelty, causality, importance, or autonomous discovery.

1 Introduction

Literature-based discovery (LBD) searches for relationships that can be inferred by connecting separate bodies of published work. Swanson’s fish-oil and Raynaud’s-syndrome case established the canonical example [1]. Subsequent studies replicated, extended, and automated such ABC-style searches [2, 3, 4]. These studies also illustrate a hard evaluation problem: a method can be adjusted after its famous target and historical outcome are already known.

Temporal link prediction offers an objective but deliberately narrower alternative. A graph is constructed at T0, a complete ordering is sealed, and edges newly observed at a later T1 are used as outcomes. Adamic–Adar is a standard local link-prediction score that discounts high-degree common neighbours [5]. This benchmark applies that family to maintained-current MeSH assignments in annual PubMed baselines. It does not claim that later co-indexing proves a new scientific relationship. The measured event is database-link emergence.

The project that hosts this benchmark, *lacuna*, previously tested two OpenAlex topic-level metrics against a pre-registered Swanson reproduction. Both failed. Those negative results remain published and were not used as evidence that the active formula works. The present protocol was frozen as a separate prospective track before its candidate scores were inspected.

2 Scope and claim boundary

The primary estimand is ranking performance for future direct MeSH co-indexing links in complete PubMed baselines. A pair’s rank does not estimate whether the relationship is true, causal, important, clinically useful, mechanistic, novel to humanity, or absent from existing knowledge. PubMed indexes academic biomedical literature and does not cover all academic work, much less craft, practitioner, indigenous, historical, humanities, or other non-academic knowledge.

The allowed positive conclusion after a pass is therefore only: the sealed formula was validated for prospective PubMed/MeSH link-emergence ranking under this protocol. “Validated knowledge-gap detector”, “autonomous scientific discovery”, and equivalent claims are outside the design.

3 Sealing chronology and materials

The machine-readable protocol first appeared on 12 August 2026. The complete T0 source, score-free candidate contract, exhaustive candidate universe, formula, and full prediction order were sealed in successive public repository commits on 13 August 2026. A source-pinned release watcher was frozen on 24 August 2026. The later registration packet records this already sealed state; it is not represented as a new preregistration.

T0 comprises 1,334 official 2026 PubMed baseline transports with 54,267,874,919 compressed bytes and 39,994,988 distinct parsed PubMed identifiers, plus the matching official MeSH XML transport with 31,110 descriptors. The official distribution locations are maintained by the U.S. National Library of Medicine [6, 7]. Every source filename, URL, byte length, SHA-256 digest, record subtotal, and descriptor-vocabulary identity is pinned. PubMed/MeSH assignments are maintained-current indexing, not period-appropriate historical indexing.

Large source corpora and computed binaries remain outside the Git repository. Small manifests pin their exact content, dimensions, and formats. Structural audits run from a clean clone; full byte-level replay additionally requires the pinned local files.

4 Score-free candidate universe

All unordered descriptor pairs (u, v) were considered in lexicographic descriptor-UI order. A pair was eligible when both endpoints occurred in at least 100 T0 articles, its independence expectation was at least 5, and its exact T0 direct co-indexing count was zero. Identical descriptors, ancestor–descendant pairs, and pairs sharing an entry term were excluded. Sampling was forbidden.

The reducer processed all 39,994,988 distinct PMIDs and measured 342,733,118 descriptor assignments and 51,128,229 distinct positive descriptor-pair rows. The resulting exhaustive stream contains 7,310,895 candidates. Candidate construction did not compute a metric, score, rank, or

prediction, and it used no human or LLM labels.

An exact-zero value here means only that no direct pair row occurred in this maintained-current T0 index. It is not evidence that the underlying relationship or knowledge is absent.

5 Frozen graph and ranking methods

Let N be the 39,994,988-record denominator, n_i the exact support of descriptor i , and c_{ij} the exact direct co-indexing count. An unweighted positive-association backbone contains edge (i, j) exactly when

$$c_{ij}N > n_i n_j. \quad (1)$$

Equality is excluded. Integer products use at least 128 unsigned bits. The sealed backbone has 31,760,211 undirected edges.

For candidate endpoints u and v , let $\Gamma(u)$ and $\Gamma(v)$ denote their backbone neighbours and $d(w)$ the unweighted degree of common neighbour w . The primary fixed-point score is

$$S_{AA}(u, v) = \sum_{w \in \Gamma(u) \cap \Gamma(v)} \left\lfloor \frac{2^{48}}{\ln d(w)} \right\rfloor. \quad (2)$$

Decimal logarithms and division use 80-digit precision with round-half-even before flooring; the sum is checked as an unsigned 64-bit integer. Missing, non-finite, or overflowing values force abstention without a prediction seal.

Five baselines were frozen with the primary formula: endpoint prevalence $n_u n_v$, common-neighbour count, resource allocation $\sum 2^{48}/d(w)$ with per-term flooring, exact-rational Jaccard similarity, and preferential attachment $d(u)d(v)$. Every ordering uses a deterministic hash-based tie key followed by the pair key, with no labels, descriptor names, or future outcomes involved.

The completed run emitted one score tuple for every eligible pair. Of 7,310,895 primary scores, 7,310,826 were nonzero. Source, formula, dependency, compiler, binary, conformance, input, output, and order hashes are sealed. These are computation-integrity measurements from an unvalidated method, not prospective results.

6 Prospective outcome acquisition

A monthly machine watcher observes only the next sequential official release identity for 2027, 2028, and 2029. It cannot overwrite an observed release. Missing, skipped, out-of-order, partial, or conflicting evidence forces abstention. API result pages are not accepted as baselines. Evaluation waits for a complete official 2029 PubMed baseline and its matching MeSH release; publication dates are reported but do not define database emergence.

A new row is a PMID–descriptor-pair row absent from T0 and present at T1. A positive outcome requires at least three distinct new co-indexed PMIDs spanning at least two journals and two publication years. A zero count is labelled only “no observed link emergence in the window”. One or two PMIDs, a retired or ambiguously mapped descriptor, failed source reconciliation, or a qualifying retraction makes the pair censored. Re-indexing of pre-existing PMIDs remains a database link event but is counted separately. Exact PMIDs and source rows must accompany every outcome.

7 Registered evaluation

The primary population contains all non-censored eligible T0 pairs. Fewer than 200 positive or 20,000 negative outcomes triggers abstention for insufficient power. Otherwise, every one of the following gates must pass:

- precision at 100 is at least 0.25;
- precision-at-100 lift over prevalence is at least 5;
- average-precision lift over prevalence is at least 3; and
- the 95% bootstrap lower bound on precision-at-100 lift is at least 2, using 10,000 replicates with the frozen seed `lacuna-autonomous-prospective-v1`.

Two automatic structural diagnostics are also gates rather than negative ground truth. Excluded immediate-parent siblings may occupy at most 1% of the primary top 1%; a deterministic degree-matched distant cohort may occupy at most 5% of the primary top 5%. The final state is pass only if all source, integrity, power, primary-metric, and structural gates pass. Any evaluable primary or structural failure is terminal for the formula. Insufficient source, integrity, outcome, or power evidence yields abstention, without manual override. A later formula requires a new protocol and new T0 seal.

8 Current status

As of 25 August 2026, the state is `predictions_sealed_waiting_for_outcome`. The verdict is `not_ready`; readiness contribution is zero; no T1 outcome has been inspected; and validated knowledge-gap pairs number zero. The sole active blocker is maturation of the three-release outcome window.

Table 1 distinguishes construction records from unavailable performance evidence.

Recorded and sealed	Not yet measured
T0 source identities and counts 7,310,895 score-free candidates	2027–2029 complete outcome window Positive, negative, and censored outcome counts
Formula, five baselines, and total orders	Precision, average precision, lift, and intervals
Integrity and refusal-to-overwrite gates	Pass, fail, or abstain evaluation verdict

Table 1: Scientific state at manuscript preparation. Sealing proves identity and computation, not prospective performance.

9 Limitations

The target is an indexing event rather than scientific truth. MeSH practices and descriptor identity can change between releases, and later indexing of old articles can create an observed link without

a newly published relationship. The protocol reports that component separately but does not remove it from the database-event target. Requiring multiple PMIDs, journals, and years reduces single-record noise but may miss real low-volume links.

Local link-prediction measures can favour structural proximity, degree patterns, and ontology artifacts. Frozen baselines and structural diagnostics expose some of this behaviour but cannot establish mechanism or utility. The metric was selected before scoring but is not independently validated. A single future pass, if obtained, would be specific to the frozen data model, outcome definition, and release window.

The project preserves negative results, generated state, and measured state with separate labels. This supports traceability and FAIR research-software practice [8], but hash-pinned availability does not by itself guarantee long-term hosting of every large source or binary file.

Data and code availability

Code, machine-readable contracts, small manifests, tests, and audit commands are available at <https://github.com/tang-vu/lacuna> under the MIT License. Software release DOI: **TO BE SUPPLIED BEFORE SUBMISSION**. Public OSF registration DOI: **TO BE SUPPLIED BEFORE SUBMISSION**. The repository records official source URLs and hashes; it does not redistribute the complete PubMed corpus.

Ethics statement

The benchmark analyzes public bibliographic metadata, recruits no participants, and makes no clinical decision. No human review, adjudication, or manual label is an active-system dependency.

Generative-AI disclosure

OpenAI Codex assisted with drafting and typesetting this manuscript on 25 August 2026. It was not used to score, rank, filter, or label T0 candidates, and it did not alter the sealed score or total order. The human author is responsible for verifying the manuscript and adapting this disclosure to the selected repository or journal policy.

Funding and competing interests

Funding: **TO BE SUPPLIED BEFORE SUBMISSION**. Competing interests: **TO BE SUPPLIED BEFORE SUBMISSION**.

References

- [1] Don R. Swanson. Fish oil, raynaud’s syndrome, and undiscovered public knowledge. *Perspectives in Biology and Medicine*, 30(1):7–18, 1986. doi:10.1353/pbm.1986.0087.

- [2] Michael D. Gordon and Robert K. Lindsay. Toward discovery support systems: A replication, re-examination, and extension of swanson’s work on literature-based discovery of a connection between raynaud’s and fish oil. *Journal of the American Society for Information Science*, 47(2):116–128, 1996. doi:10.1002/(SICI)1097-4571(199602)47:2<116::AID-ASI3>3.0.CO;2-1.
- [3] Marc Weeber, Henny Klein, Lolkje T. W. de Jong-van den Berg, and Rein Vos. Using concepts in literature-based discovery: Simulating swanson’s raynaud–fish oil and migraine–magnesium discoveries. *Journal of the American Society for Information Science and Technology*, 52(7):548–557, 2001. doi:10.1002/asi.1104.
- [4] Judita Preiss and Mark Stevenson. Quantifying and filtering knowledge generated by literature based discovery. *BMC Bioinformatics*, 18(Suppl 7):249, 2017. doi:10.1186/s12859-017-1641-9.
- [5] Lada A. Adamic and Eytan Adar. Friends and neighbors on the web. *Social Networks*, 25(3):211–230, 2003. doi:10.1016/S0378-8733(03)00009-1.
- [6] U.S. National Library of Medicine. MEDLINE/PubMed data documentation. https://www.nlm.nih.gov/databases/download/pubmed_medline.html, 2026. Accessed 2026-08-25.
- [7] U.S. National Library of Medicine. MeSH xml data elements. https://www.nlm.nih.gov/mesh/xml_data_elements.html, 2026. Accessed 2026-08-25.
- [8] Michelle Barker, Neil P. Chue Hong, Daniel S. Katz, Anna-Lena Lamprecht, Carlos Martinez-Ortiz, Fotis Psomopoulos, Jennifer Harrow, Leyla Jael Castro, Morane Gruenpeter, Paula Andrea Martinez, and Tom Honeyman. Introducing the FAIR principles for research software. *Scientific Data*, 9(1):622, 2022. doi:10.1038/s41597-022-01710-x.